Embedding Environmental Management in High-Polluting Firms: The Role of Green Digital Policy and Collaborative Innovation

Abstract: Against the backdrop of intensifying institutional pressure for green transformation and the strategic rollout of digital infrastructure, whether green digital policies can effectively incentivize high-polluting firms to engage in green collaborative innovation remains a question requiring micro-level causal identification. This study exploits a quasi-natural experiment based on China's national green data center pilot program and a panel dataset of heavily polluting firms listed on the Shanghai and Shenzhen A-share markets from 2007 to 2023. The results show that green digital policy significantly enhances green collaborative innovation among high-polluting firms, whereas no comparable effect is observed in non-polluting firms. Mechanism analysis reveals that the policy operates through three pathways: strengthening data asset disclosure, promoting integration into global innovation networks, and mitigating risks associated with green innovation failure. Further moderation analysis suggests that both data element utilization and green strategic orientation significantly reinforce the policy effect, while industry-level green competition intensity exerts a dampening influence. Moreover, the policy stimulates both the extensive and intensive margins of green cooperation and exerts significant effects on both green invention and utility model patents. These findings highlight the institutional functions of green digital policy in embedding firm capabilities, alleviating innovation risk, and expanding collaborative boundaries, offering empirical support for evaluating its performance and optimizing green governance strategies for high-pollution firms.

Keywords: Green Digital Policy; Green Collaborative Innovation; High-Polluting Firms; Data Asset Disclosure; Global Innovation Networks

1. Introduction

Guided by China's dual carbon objectives, the national strategy for high - quality development has emphasized a dual - track paradigm that seamlessly integrates green transformation with digital infrastructure development (Dogan et al., 2025 ;Nizam et al., 2020). The report of the 20th National Congress emphasized accelerating the building of a strong manufacturing sector, a high-quality economy, and a digital China, guiding industrial development toward higher-end, intelligent, and low-carbon pathways. President Xi Jinping has repeatedly called for the coordinated development of digitalization and greenization to facilitate structural upgrades in energy consumption, production, and lifestyles, aiming to foster new advantages in ecological civilization (Hu et al., 2023; Zhang et al., 2022). Under this strategic orientation, green low-carbon development and new digital infrastructure have become parallel national priorities, with institutional arrangements and policy implementation proceeding in a coordinated and mutually reinforcing manner.

Consequently, recent policy directives have mandated the profound integration of digitalization and green development across diverse sectors. These policy documents champion the extensive deployment of cloud computing, data - driven applications, and intelligent systems, while urging firms to harness both digital and green technologies to modernize traditional industrial processes (Cvjetko, 2015; Hao et al., 2023; Lee et al., 2023). As a representative institutional arrangement, green digital policies (GDPs) reflect the convergence of environmental governance and digital transformation. For example, in 2015, the Ministry of Industry and Information Technology (MIIT), in conjunction with other central agencies, launched the "National Green Data Center Pilot Program," which mandated improvements in energy efficiency and the promotion of green, intensive development of data infrastructure. This convergence of digital infrastructure strategy and green policy across both temporal and spatial dimensions has laid an institutional foundation for understanding the synergistic effects of GDPs.

At the industry level, heavily polluting firms (HPFs) characterized by large-scale pollutant emissions are critical governance targets for green transformation (Shaikh 2017; Lin & Pan, 2024; Yang et al., 2024). These firms demonstrate significant marginal potential for pollution mitigation, and fostering their participation in green innovation endeavors can yield substantial environmental benefits. Concurrently, the escalation of public environmental consciousness and the formal institutionalization of carbon neutrality objectives have imposed mounting societal and regulatory pressures on HPFs (Gao et al., 2022; Liu et al., 2023). Many HPFs have begun to embrace digital technologies as part of their transition toward green innovation, seeking to reconcile environmental protection with economic performance (Zheng & Zhang, 2023; Yang et al., 2023; Zhao et al., 2023). Given their dual role as major emitters and key agents of transformation, understanding the determinants of green collaborative innovation within HPFs is of both theoretical and practical relevance to advancing industrial green upgrading.

Prior research has explored the determinants of corporate green innovation from a multitude of perspectives. Environmental regulation has consistently been acknowledged as a pivotal focal point within the extant literature on green technological innovation (Lv et al., 2021; Du et al., 2021; Yang et al., 2025a). A large body of empirical research demonstrates that mandatory environmental laws and market-based incentives significantly stimulate firms to engage in green technology investment and innovation (Li & Gao, 2022; Wang et al., 2022; Zhou et al., 2023). For instance, evidence shows that environmental regulation and market competition jointly serve as key motivators for corporate green investment decisions, shaping firms' innovation behavior in complementary ways (Wang et al., 2022; Li et al., 2023; Peng et al., 2021). Moreover, stakeholder-related factors such as media attention and government subsidies have also been shown to exert positive influence on firms' green strategic choices, providing additional impetus for green investment (He et al., 2022; Li et al., 2023; Hu et al., 2021). These studies highlight the complex and multifaceted nature of green innovation under institutional pressure and stakeholder scrutiny.

In the context of digital transformation, researchers have increasingly focused on the enabling effects of digital technology on green innovation. Digital transformation has been empirically demonstrated to substantially elevate green innovation efficiency, augment both the quality and quantity of innovation outputs, and facilitate the attainment of long - term sustainability objectives (Xiao et al., 2024; Yin et al., 2022). Subsequent research has unveiled that digitalization underpins green collaborative innovation by enhancing information transparency, optimizing resource allocation, and fortifying technological integration (Melander et al., 2019; Shao et al., 2024; Cammarano et al., 2024; Qamri et al., 2025). In other words, within the digital economy, firms are not only better able to optimize internal green innovation processes but also more capable of engaging in inter-organizational collaboration with supply chains, research institutions, and other external stakeholders to jointly navigate the challenges of low-carbon transition (Yang et al., 2025b). While the literature affirms the mutually reinforcing relationship between digitalization and greening, the synergy effects remain context-dependent and heterogeneous, calling for more systematic investigation into their mechanisms.

Despite the growing research on digitalization and green innovation, few studies have directly explored how GDPs affect green collaborative innovation in HPFs. In particular, under the combined momentum of the dual carbon targets and the national digital China strategy, theoretical inquiry into how institutionalized green digital policies incentivize green innovation in highly polluting sectors remains underdeveloped. This study addresses this gap by taking GDPs especially the National Green Data Center pilot—as a representative institutional setting and systematically analyzing their effects on green collaborative innovation in HPFs. By integrating the perspectives of environmental regulation and digital enablement, the study contributes to the theoretical synthesis of green innovation and digital economy research, while offering policy-relevant insights for industrial green transformation in pollution-intensive sectors.

This paper makes the following noteworthy contributions to the academic literature. First, whereas the extant research on GDPs and green innovation has predominantly focused on unidimensional incentive channels, with scant attention paid to the structural effects of policy integration, this study offers a novel conceptualization. It treats GDPs as an institutional convergence point between environmental governance and digital infrastructure, and delves into how these policies reshape firms' cognitive and resource boundaries, thereby influencing green collaborative innovation within HPFs. This approach not only broadens the scope of environmental regulation theory but also provides valuable insights for the design of precision - oriented green governance institutions. Second, in stark contrast to previous studies that have centered on generic firms or broad industry samples, this paper explicitly takes HPFs as the unit of analysis. It uncovers the institutional response mechanisms and the heterogeneity in green innovation behaviors among high - polluting entities. The findings underscore that the effectiveness of GDPs is contingent on three organizational capability dimensions: data asset disclosure, global innovation network embeddedness, and innovation risk management. This contributes a well - structured explanation of how institutional environments interact with firm capabilities to shape innovation outcomes, thereby filling a critical gap in the existing literature. Third, by elucidating the micro - level mechanisms through which firms respond to GDPs namely, data governance, risk buffering, and international collaboration this study constructs a coherent theoretical chain that links policy intervention, organizational capability, and collaborative innovation. This mechanism - based perspective enhances the theoretical integration of digital and green transformation and offers empirical support for policy tool selection at the intersection of institutional design and firm - level adaptation, thus providing a solid foundation for future research and policy - making in this area.

2. Theoretical Framework and Hypotheses

2.1 GDPs and Green Collaborative Innovation in HPFs

Amid a concerted national drive for green transformation and digital infrastructure development, GDPs have emerged as pivotal institutional catalysts for corporate green innovation, particularly within environmentally intensive industries where their incentive effects exhibit a structurally more pronounced impact (Li et al., 2025; Guo et al., 2023; Qamri et al., 2022;Zhang etal.,2025). For HPFs, green collaborative innovation serves not only as a mechanism to alleviate regulatory compliance pressure, but also as an organizational strategy for overcoming internal capability constraints and reconfiguring technological trajectories. Whether institutional incentives can be effectively translated into inter-organizational collaborative actions hinges on how firms interpret and internalize policy signals, allocate resources accordingly, redefine risk boundaries, and connect with external innovation networks.

By institutionalizing data infrastructure, GDPs transform the external environment in which firm’s access, perceive, and respond to opportunities for collaborative green innovation. Critically, this institutional effect does not hinge on direct resource allocation but rather operates through an indirect reconfiguration of organizational cognition and behavioral boundaries. On one hand, the policy-induced standardization of data governance improves the accessibility and granularity of green information, reducing the search and matching costs of collaboration and expanding firms’ perceptual boundaries for identifying potential green partners. On the other hand, institutionalized green digital infrastructure reconfigures technical resource networks, enabling otherwise isolated HPFs to access more interconnected collaborative channels. This shift facilitates their integration into broader innovation ecosystems via structural realignment.

In such firms, the key to behavioral responsiveness lies in the behavioral stability that GDPs offer. Under high uncertainty, green collaborative innovation is often constrained by elevated risks of failure, deep interdependence of capability structures, and ambiguity in outcome attribution. The institutional safeguards embedded in GDPs help mitigate these structural barriers, thus incentivizing firms to pursue external cooperation for technological breakthroughs (Arfi et al., 2018; Reficco et al., 2018). A stable stream of institutional signals enhances organizational confidence and tolerance toward boundary-spanning green innovation initiatives.

Taken together, by reshaping informational environments, restructuring collaborative architectures, and stabilizing behavioral expectations, GDPs are likely to promote green collaborative innovation in HPFs:

Hypothesis 1: GDPs significantly promote green collaborative innovation among HPFs.

2.2 Mechanism: Data Asset Disclosure Capability

Under the influence of GDPs, the degree to which firms can translate institutional signals into actionable collaborative innovation fundamentally hinges on their capacity to identify, structure, and communicate data elements. As quasi-institutional resources embedded within firms, data assets not only reflect internal information governance capabilities but also constitute the primary interface through which firms achieve cognitive alignment with external institutional environments. Unlike traditional disclosures focused on financial or governance attributes, data asset disclosure is marked by structural ambiguity and definitional fluidity. Consequently, a firm's ability to present such assets in a comprehensible and credible manner is pivotal in determining whether institutional provisions can be effectively internalized.

Green collaborative innovation, by nature, is a highly complex, boundary-spanning activity. Its realization presupposes that firms possess sufficient informational expressiveness to reduce the cognitive barriers of potential partners (Melander et al., 2017; Melander & Arvidsson, 2022). Under GDPs, firms with stronger data disclosure capabilities are more likely to have their green practices, resource endowments, and technical competencies recognized externally, thereby increasing their likelihood of being incorporated into collaborative innovation networks. The transmission of institutional incentives is not automatic; rather, it depends on whether a firm can construct a data architecture that renders it identifiable and credible as a potential collaborator an architecture that centers around data assets.

Furthermore, the capacity for data asset disclosure underscores a firm's organizational agility in navigating institutional complexity. Within rapidly evolving green policy landscapes characterized by heterogeneous standards and dynamic signaling, firms must establish robust disclosure frameworks that facilitate continuous alignment with policy trajectories. Those enterprises capable of systematically and promptly signaling their green credentials and digital competencies demonstrate superior institutional adaptability and programmable compatibility, thereby enhancing their likelihood of being leveraged by policy windows and receiving favorable institutional feedback.

Therefore, data asset disclosure capability serves not only as an observable manifestation of institutional receptivity but also as a necessary condition for network-level collaboration. Under

GDPs, firms with more advanced disclosure capabilities are better positioned to leverage institutional stimuli to form cross-organizational green partnerships:

Hypothesis 2: The positive effect of GDPs on firms’ green collaborative innovation is significantly stronger for firms with greater data asset disclosure capability.

2.3 Mechanism: Embeddedness in Global Innovation Networks

Green collaborative innovation demands the reconfiguration of firms' cognitive and resource architectures across organizational boundaries a particularly formidable challenge for HPFs already grappling with transformational pressures. In this context, the efficacy of GDPs in fostering collaborative innovation depends critically on whether firms maintain adequate structural linkages to external environments, particularly their degree of embeddedness within global innovation networks. Such embeddedness not only enhances firms’ capacity to access external knowledge but also determines their ability to position themselves within complex technological fields, mobilize heterogeneous resources, and engage in coordinated action.

GDPs, by increasing institutional signal density and improving the interconnectivity of technical infrastructure, offer policy windows through which firms can reconfigure their transnational innovation ties. For firms already embedded in global innovation networks, institutional signals no longer function merely as compliance constraints; instead, they serve as coordination anchors that facilitate the stabilization of cross-border collaborations in uncertain policy environments (Li et al.2024; Meyer et al., 2023; Bradley et al., 2021). In contrast, firms lacking global structural linkages tend to rely more heavily on internal capabilities or local policy arrangements, making it difficult for them to expand their collaborative boundaries in response to GDPs.

The organizational logic of global innovation networks makes it more likely for firms to view green collaboration as an optimal strategy for resource orchestration and knowledge integration. Highly embedded firms tend to possess richer external search capabilities, more stable institutional bridging structures, and more mature mechanisms for governing collaboration. These features enable them to respond to policy incentives with lower levels of organizational friction. The institutional stability and infrastructural connectivity introduced by GDPs amplify the capacity of such firms to leverage collaborative spaces more effectively.

From an institutional perspective, GDPs are not uniformly effective across all firms. Their impact is contingent upon the existence of organizational structures that enable cross-firm coordination. In this transmission process, global innovation network embeddedness emerges as a key mediating mechanism:

Hypothesis 3: The positive effect of GDPs on green collaborative innovation is more pronounced among firms with higher levels of embeddedness in global innovation networks.

2.4 Mechanism: Risk of Innovation Failure

Green collaborative innovation is inherently a highly uncertain techno-organizational co-creation process. Its success depends not only on firms' willingness to invest or the availability of resources, but more fundamentally on how firms perceive the consequences of failure and the degree of organizational tolerance they maintain toward such outcomes (Cricelli et al., 2023; Forsman, 2021). This is especially true for HPFs, where green transition typically involves high costs, low certainty, and complex regulatory pressures. These factors heighten firms' sensitivity to innovation failure and may inhibit their motivation to initiate collaborative ventures. For GDPs to translate into substantive collaborative outcomes, they must pierce through these risk-averse structures and offer more controllable mechanisms for coping with failure.

The institutionalization of digital infrastructure under GDPs reshapes the expectation structure surrounding innovation failure. On one hand, policy pilots enhance transparency in resource allocation and standardization of governance procedures, alleviating firms' fears that innovation failure will automatically trigger external sanctions. This, in turn, incentivizes firms to treat green innovation as a strategic commitment rather than symbolic compliance. On the other hand, improved information integration under digital policy frameworks strengthens firms' capacity to monitor potential disruptions, incompatibilities, or failures within collaborative projects, thus reducing delayed reactions to emerging risks.

More critically, GDPs establish external risk-buffering mechanisms by leveraging institutional infrastructure. For firms with limited prior exposure to green governance frameworks or underdeveloped organizational capabilities, these institutional arrangements partially restructure the attribution of failure and enhance firms' tolerance for uncertainty. Rather than fully internalizing green innovation risks, firms can now rely on institutional scaffolding that improves outcome predictability (Zhao et al., 2025). This mechanism substantially elevates firms' propensity to engage in cross-organizational green collaboration, transforming it from a perceived liability into a viable strategic option. Accordingly, under the influence of GDPs, firms with lower levels of green innovation failure risk are more likely to translate institutional incentives into substantive collaborative behavior:

Hypothesis 4: The positive effect of GDPs on green collaborative innovation is more pronounced among firms with lower levels of innovation failure risk.

3. Research Design

3.1 Identification of HPFs and Sample Construction

This study uses A-share listed firms in Shanghai and Shenzhen from 2007 to 2023 as the baseline sample, focusing on HPFs firms most constrained by green transition mandates and most responsive to institutional incentives. Following the classification criteria proposed by Guo et al. (2019), and based on the Ministry of Environmental Protection's catalog of 14 high-environmental-risk industries, we define firms in industries with codes B06–B12, C13, C15, C17, C19, C22, C25–C32, D44, and D45 as representative of heavily polluting sectors. Firms were then screened accordingly.

To ensure comparability in operational status and financial structure, we exclude firms labeled ST or *ST due to major financial distress. All continuous variables were winsorized at the 1st and 99th percentiles to mitigate the influence of outliers. The final sample consists of 13,826 firm-year observations. All financial and governance data are sourced from the CSMAR database.

3.2 Variable Construction

1. Dependent Variable.

The dependent variable is green collaborative innovation (Greenco), which measures the extent to which HPFs engage in cooperative green innovation activities with external entities. Following the approach of Yang and Wang (2024), we measure this by counting the total number of green patents jointly filed by listed firms and other organizations such as enterprises, universities, and research institutions.

Specifically, we first use the green patent classification list provided by the World Intellectual Property Organization to search the China National Intellectual Property Administration (CNIPA) database for green patents jointly filed by A-share listed firms and other entities. To improve data accuracy, unmatched entries were subjected to fuzzy matching using Python's FuzzyWuzzy library. Results with a matching score ≥ 80 were manually verified to eliminate mismatches arising from name changes or incomplete firm information. Additionally, for applicants with generic descriptors such as "company," "plant," or "research institute," further validation was performed to ensure correctness.

Finally, for each listed firm and year, we calculate the total number of jointly filed green patents. To mitigate skewness in distribution, we take the natural logarithm of one plus this count, denoted as:$Greenco_{it} = \ln(Joint\ Green\ Patents_{it} + 1)$

2. Independent Variable.

The core policy examined in this study is the national green data center pilot program, which serves as the institutional embodiment of GDPs. Launched in 2015 by the Ministry of Industry and Information Technology (MIIT), the National Development and Reform Commission (NDRC), and other central agencies, the program was institutionalized in 2018 and has since been implemented annually. It targets energy-intensive sectors such as telecommunications, internet services, and finance, aiming to promote the green transformation of digital infrastructure by introducing energy efficiency evaluations, green technology catalogs, and fiscal incentives. Eligibility for pilot status is centrally administered and subject to provincial-level recommendation and third-party evaluation, rendering the program a non-market-based institutional selection mechanism with quasi-exogenous characteristics. We identify treatment status using policy announcements from 2018 to 2023, determining the first year each city was approved as the baseline for policy implementation. Based on this staggered rollout, we construct a multi-period difference-in-differences (DID) specification. Specifically, Treat is a binary variable equal to 1 if the firm is located in a pilot city; Post equals 1 in years following the city's pilot approval; the interaction term DID = Treat × Post captures whether a firm is exposed to the policy during its active implementation window. This variable exploits the temporal and spatial variation in the rollout to form a quasi-natural experiment, serving as the core independent variable for identifying the causal effects of GDPs.

3. Control Variables

Firm size (Size) is measured as the logarithm of total assets and reflects the firm's resource endowment and scale externalities. Leverage (Lev) captures the constraints of capital structure on risk preferences. Return on assets (ROA) is defined as net income over total assets and reflects operating performance and internal financing capability. Operating cash flow intensity (Cashflow) is defined as cash flow from operating activities divided by total assets and captures the degree of liquidity constraint. Listing age (ListAge) is measured as the difference between the observation year and the firm's IPO year, controlling for organizational maturity and stability in external governance. CEO duality (Dual) is a binary variable equal to 1 if the board chair and the CEO are the same person, reflecting top-level governance structure.

4. Mechanism Variables.

To identify the internal channels through which green digital policies affect collaborative innovation, this study constructs three mechanism variables, corresponding to global collaboration capacity, information transparency, and innovation failure probability.

The variable GlInNet indicates whether a firm is embedded in global innovation networks, following Li et al. (2022), and captures its capacity to form transnational knowledge linkages. Based on the "Overseas Affiliate Companies" table in the CSMAR database, if a firm establishes a foreign subsidiary in a given year, and the subsidiary is not registered in offshore financial centers (such as Hong Kong, the British Virgin Islands, the Cayman Islands, Jersey, etc.), then the firm is considered embedded in global innovation networks from that year onward and is assigned a value of 1; otherwise, the value is 0. Multinational subsidiaries, as the "institutionalized interface" for enterprises to acquire overseas technology, market and management experience, can promote cross-border knowledge transfer. This indicator directly reflects an enterprise's ability to build cross-border knowledge links through its physical presence. Secondly, the core of the global innovation network is the continuous interaction among entities (Owen-Smith & Powell, 2004). The establishment of overseas subsidiaries indicates that the enterprise has invested resources to build long-term overseas bases, which better reflects deep integration than short-term cooperation or trade relations.

The variable DataDisclosure reflects a firm's governance capability in defining, identifying, and communicating data elements, representing its transparency in collaborative efforts under institutional conditions. The construction of this variable follows the method of Niu et al. (2024), based on keywords listed in the Data Asset Management White Paper (Version 4.0) by the China Academy of Information and Communications Technology. A disclosure dictionary is built using Word2Vec semantic expansion, and after filtering out noise terms, the frequency of valid disclosure keywords is calculated as a proportion of total word count in the Management Discussion and Analysis (MD&A) section to construct an annual disclosure intensity indicator.

The variable InnoRisk captures the uncertainty level of green R&D and follows the approach of Wang et al. (2019). It is defined as a binary variable: if a firm's growth rate of R&D expenditure in a given year exceeds the growth rate of net profit in the subsequent year, it is deemed to have high innovation failure risk and is assigned a value of 1; otherwise, the value is 0.

5. Moderating Variables.

The effects of green digital policies on green collaborative innovation are not homogeneous. Their marginal effects are contingent upon firms' internal digital capabilities, strategic cognitive structures, and the degree of institutional competition within their industries. To identify this heterogeneity, we introduce three moderating variables reflecting firm-specific attributes and industry environments.

The first type of moderator concerns a firm's actual capacity to utilize data elements. Following Shi et al. (2023), we calculate the annual frequency of five keyword categories artificial intelligence, blockchain, cloud computing, big data technologies, and big data application disclosed in annual reports. We then construct a composite index of data element utilization level (DataUtil) to reflect a firm's absorptive and transformative capacity for data resources. This indicator captures not only the degree of digital infrastructure integration but also the firm's ability to identify collaborative green opportunities embedded within data assets.

The second moderator focuses on whether a firm has a clearly articulated green strategic orientation. Drawing on Yang (2024), we extract seven indicators from annual reports and CSR disclosures: whether the firm articulates environmental protection concepts, sets environmental targets, establishes an environmental management system, obtains ISO14000 certification, provides environmental training, engages in environmental campaigns, and implements the "three-simultaneity" system. The total score of these indicators constitutes the firm's green strategic orientation (GreenStrategy). This variable aims to capture whether green innovation is embedded in the firm's long-term strategic roadmap rather than representing an ad hoc response to policy signals.

The third moderator reflects the institutional context of the firm's industry. Following Zheng et al. (2025), we measure industry-level green competition intensity (IndGreenComp) as the proportion of firms within the same industry that have obtained ISO14001 environmental management system certification. This serves as a proxy for the strength of external institutional pressure. The extent of industry-wide green regulation is expected to shape how firms respond to GDPs and define the strategic boundary of their responses.

3.3 Model Specification

The core advantage of high-dimensional fixed-effect models lies in their powerful ability to control heterogeneity. By introducing multi-dimensional fixed effects such as individuals, time, industries and regions, this model can effectively capture unobtrusive heterogeneous characteristics, thereby alleviating the problem of omitted variable bias. To discern the impact of green digital policies on green collaborative innovation among HPFs, we employ a high - dimensional fixed - effects model, which is specified as follows:

$$\text{Greenco}_{it} = \beta \cdot \text{Policy}_{it} + \gamma X_{it} + \mu_i + \delta_c + \lambda_t + \varepsilon_{it}$$

where Greenco_{it} denotes the level of green collaborative innovation for firm i in year t, and Policy_{it} is the core explanatory variable indicating whether the firm is under active policy coverage of the green data center initiative. The coefficient β captures the average treatment effect. The control vector X_{it} includes contemporaneous financial and governance characteristics such as firm size, leverage, profitability, liquidity, listing age, and managerial structure. The model absorbs firm fixed effects μ_i, city fixed effects δ_c, and year fixed effects λ_t, with standard errors clustered at the firm level.

To explore the internal mechanisms through which the policy operates, we augment the baseline model by including mediating variables M_{it}, resulting in the following specification:

$$\text{Greenco}_{it} = \beta' \cdot \text{Policy}_{it} + \theta \cdot M_{it} + \gamma' X_{it} + \mu_i + \delta_c + \lambda_t + \varepsilon'_{it}$$

where M_{it} represents three firm-level channels: data asset disclosure capability (DataDisclosure), embeddedness in global innovation networks (GlInNet), and green innovation failure risk (InnoRisk). The coefficient θ captures the strength of the policy's transmission through these mechanisms. All other specifications remain identical to the baseline model.

To assess whether the policy effects vary systematically with firm-specific heterogeneity, we further introduce moderating variables Z_{it} and construct the following interaction model:

$$\text{Greenco}_{it} = \beta_1 \cdot \text{Policy}_{it} + \beta_2 \cdot Z_{it} + \beta_3 \cdot (\text{Policy}_{it} \times Z_{it}) + \gamma X_{it} + \mu_i + \delta_c + \lambda_t + \varepsilon_{it}$$

where Z_{it} denotes the firm-level moderator, with three variables considered sequentially: DataUtil, GreenStrategy, and IndGreenComp. The interaction coefficient β_3 captures the marginal variation in the policy effect conditional on firm characteristics and reflects the differentiated incentive effects under heterogeneous structures. All other specifications are consistent with the baseline model.

4. Empirical Results and Analysis

4.1 Descriptive Statistics

Prior to the implementation of green digital policies, firms' green collaborative innovation capabilities exhibited structurally uneven distributions. As shown in Table 1, among HPFs, those located in pilot cities reported significantly fewer collaborative green patents than their counterparts in non-pilot areas, with means of 0.042 versus 0.066, respectively (diff = 0.024, $p < 0.01$). This lack of collaborative innovation provides both a practical rationale for institutional intervention via green digital infrastructure and latent scope for the policy to generate marginal effects.

There also exist limited but statistically significant differences between the two groups in governance and organizational characteristics. Firms in the treatment group tend to be slightly smaller in scale (diff = 0.289, $p < 0.01$), exhibit a higher proportion of CEO duality, and have shorter listing histories (diff = 0.126, $p < 0.01$), all of which suggest greater sensitivity to novel institutional signals. In contrast, no material differences are observed in leverage (Lev), return on assets (ROA), or operating cash flow intensity between the two groups, indicating a high degree of comparability in financial fundamentals. This combination moderate heterogeneity in organizational structure but strong alignment in financial conditions provides credible support for causal identification using a difference-in-differences (DID) framework.

Table 1. Descriptive Statistics: Treated vs. Control Firms in Polluting Industries

Variable	Mean_Control	Mean_Treated	Difference	Std_Err	P_value
Greenco	0.066	0.042	0.024	0.005	0.000
Size	22.244	21.955	0.289	0.022	0.000
Lev	0.409	0.408	0.001	0.004	0.877
ROA	0.048	0.048	0.000	0.001	0.949
Cashflow	0.060	0.059	0.002	0.001	0.174
ListAge	2.146	2.021	0.126	0.017	0.000
Dual	0.240	0.268	-0.028	0.008	0.000

Note: This table reports pre-treatment means of key firm-level variables for treated and control firms in polluting industries. "Treated" refers to firms located in cities included in the green data center pilot program. "Greenco" denotes the level of green collaborative innovation, measured by the log count of green co-patents filed jointly with external entities. The last three columns report the mean difference, standard error, and the associated p-value from a two-sample t-test. Firm size is measured by the natural logarithm of total assets. "Dual" is an indicator for CEO–Chair duality. All monetary variables are winsorized at the 1% level. Differences significant at the 1% level are denoted as statistically meaningful.

Figure 1 delineates the temporal evolution of green collaborative innovation among HPFs across the pre- and post-implementation periods of the green data center policy. Prior to the policy intervention, the treatment group systematically demonstrated lower levels of collaborative green innovation relative to the control cohort, with this disparity further exacerbating post-2012 a pattern indicative of enduring structural deficits in collaborative green technological capabilities. This observed trajectory aligns with the descriptive evidence presented earlier and underscores the existence of pre-policy margins for potential efficiency gains in innovation collaboration.

Starting in 2018, the treatment group experienced a marked increase in green collaborative innovation, and from 2020 onward, their performance consistently surpassed that of the control group—indicating a reversal in relative positioning. This upward shift closely coincides with the nationwide expansion of policy coverage and offers preliminary evidence of the policy's incentivizing effect on firms' green collaboration capabilities. While the figure itself does not directly identify causal effects, the observed trend aligns with the timing of the estimated treatment effects in subsequent regressions and thus offers visual support for the identification strategy.

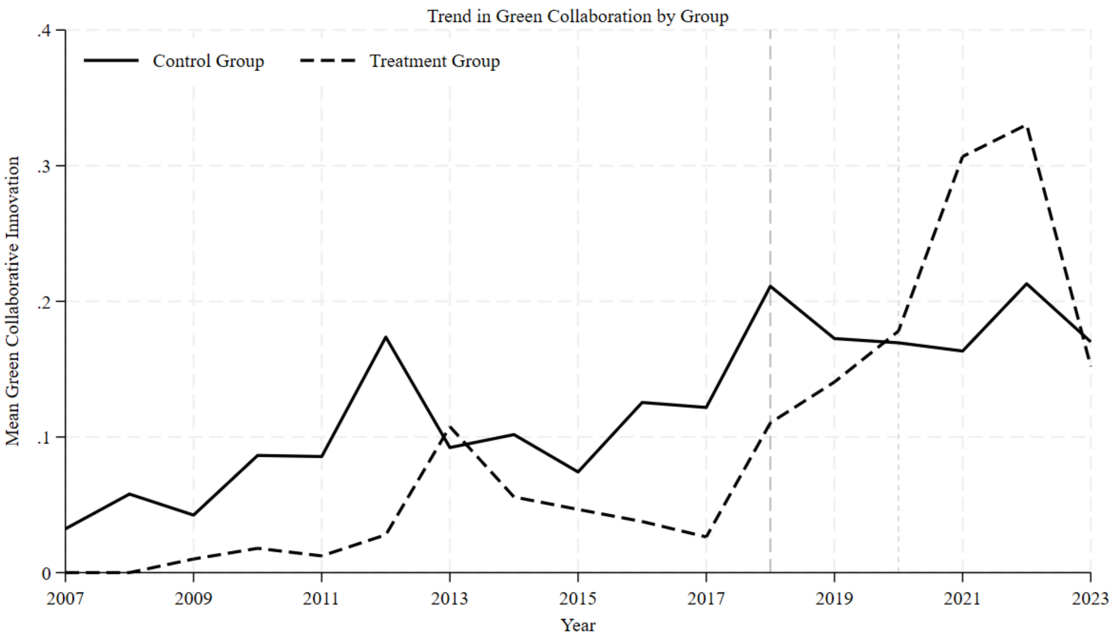

Figure 1. Trends in Green Collaborative Innovation: Treated vs. Control Firms

Note: This figure plots the average level of green collaborative innovation for treated and control firms in polluting industries from 2007 to 2023. Green collaboration is measured by the log-transformed count of jointly filed green patents. Treated firms are those located in cities included in the national green data center pilot program. Dashed vertical lines mark major policy expansion years (2018, 2020–2023). The treatment group exhibits persistently lower innovation prior to policy implementation and a sharp reversal thereafter, consistent with dynamic treatment effects.

4.2 Baseline Regression Results

Table 2 reports the estimation results examining the impact of green digital policies on the level of green collaborative innovation among HPFs. Column (1) incorporates a comprehensive set of controls, including firm-level financial and governance characteristics, as well as city-year fixed effects to account for unobserved heterogeneity. The estimated coefficient on the policy variable is both positive and statistically significant at conventional levels (coefficient = 0.076, SE = 0.022), providing evidence of a robust and economically meaningful effect of the policy intervention on collaborative green innovation activities. The above results are relatively consistent with the research of Li (2025) and Guo & Xu (2024), that is, the green attribute inherent in digital technology itself has effectively promoted the green development of enterprises.

After adding firm fixed effects in column (2), the estimated coefficient slightly declines but remains statistically significant (coefficient = 0.071, SE = 0.029, $p < 0.05$), indicating that the results are not driven by time-invariant firm heterogeneity. Column (3) further absorbs firm, city, and year fixed effects simultaneously, and the effect remains stable and highly significant (coefficient = 0.082, SE = 0.030, $p < 0.01$), reinforcing the robustness of the treatment effect under a rich set of controls for unobservable confounders.

In contrast, column (4) replaces the sample with non-HPFs. The estimated coefficient substantially decreases (coefficient = 0.004, SE = 0.014) and loses statistical significance. This finding suggests that the marginal effect of the green data center policy is concentrated among firms facing stronger green transition constraints and exhibiting higher institutional responsiveness. The result further supports the interpretation that the policy is structurally aligned with its intended target group.

Table 2. Baseline Estimates: Effect of Green Data Center Policy on Green Collaboration

Variable	(1) Greenco	(2) Greenco	(3) Greenco	(4) Greenco
Policy	0.076***	0.071**	0.082***	0.004
	(0.022)	(0.029)	(0.030)	(0.014)
Controls	Yes	Yes	Yes	Yes
Firm FE	No	Yes	Yes	Yes
City FE	Yes	No	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	13,792	13,784	13,754	31,567
Adj.R2	0.114	0.291	0.286	0.537

Note: This table reports difference-in-differences estimates of the impact of the green data center pilot policy on green collaborative innovation. The dependent variable is the log count of jointly filed green patents between the firm and external entities. “Policy” is a treatment indicator equal to one if the firm is located in a pilot city and observed in or after the treatment year. All columns include firm-level controls: log total assets (Size), leverage (Lev), return on assets (ROA), cash flow ratio, listing age, and an indicator for CEO–Chair duality. Standard errors are clustered at the firm level. Column (4) uses a non-polluting firm sample as a placebo test. *** $p < 0.01$, ** $p < 0.05$.

4.3 Robustness Checks

1. Parallel Trend Test

Figure 2 presents dynamic treatment effect estimates based on an event-study framework, employed to test the pre-policy parallel trend assumption. The left panel adopts a conventional event-time specification using policy-period interaction terms. The coefficients for the ten pre-treatment periods remain close to zero, and most confidence intervals include the zero line—providing strong support for the validity of the parallel trend assumption.

The right panel implements the cohort-stacking approach proposed by Cengiz et al. (2019). Results are consistent: prior to treatment, there is no systematic divergence in green collaborative innovation between treated and untreated groups. This reinforces the credibility of the identification strategy.

Post-treatment dynamics indicate a sharp and sustained rise in the estimated effects, with the treatment effect peaking in the third year (coefficient $\approx$ 0.25). The trajectory suggests that the policy's impact on collaborative green innovation is both cumulative and lagged in nature. The two estimation strategies yield qualitatively similar results in both direction and timing of the treatment effect, lending robustness to the baseline findings and ruling out spurious short-term fluctuations as the primary driver of the average treatment effect.

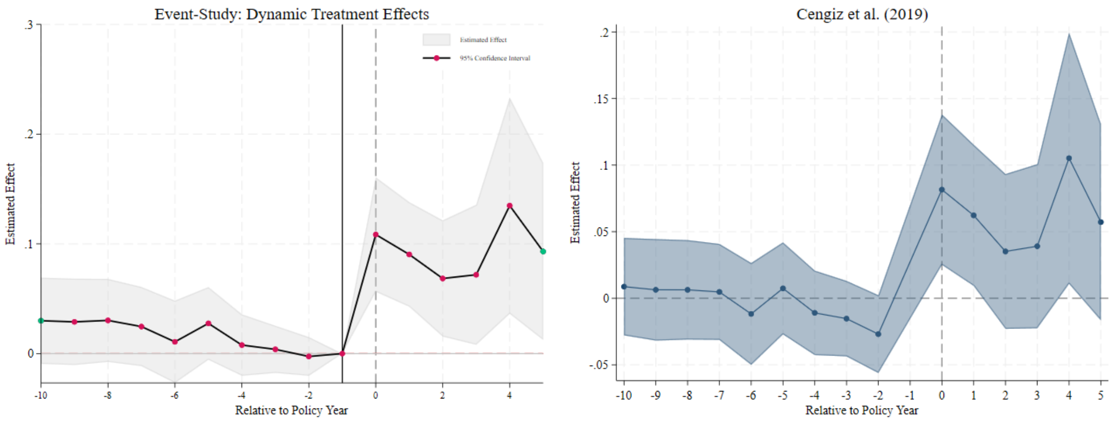

Figure 2. Event-Study Estimates of the Policy Effect on Green Collaborative Innovation

Note: This figure presents dynamic treatment effect estimates of the green data center policy using two specifications. The left panel shows coefficient plots from a traditional event-study with leads and lags of the treatment indicator. The right panel follows the cohort-based estimation approach proposed by Cengiz et al. (2019). In both cases, pre-treatment estimates are statistically indistinguishable from zero, supporting the parallel trends assumption. Post-treatment effects are positive, increasing over time and peaking around the third year after initial exposure. Shaded areas represent 90% confidence intervals.

Estimates control for firm characteristics and absorb firm, city, and year fixed effects. Standard errors are clustered at the firm level.

2. Placebo Test

Figure 3 reports the results of a placebo test based on artificially constructed treatment groups. The purpose is to assess whether the observed treatment effects may stem from spurious sample structures or time-trending errors rather than the policy intervention itself. Specifically, multiple sets of pseudo-treated firms and fake treatment years were randomly generated. For each simulation, the treatment coefficient was re-estimated, and the joint distribution of the resulting estimates and corresponding p-values was plotted.

The simulation results show that the vast majority of placebo treatment estimates cluster around zero, with associated p-values exceeding 0.1—indicating an absence of significant effects under random assignment. In contrast, the actual policy estimate (vertical line at β ≈ 0.08) lies in the right tail of the simulated distribution, clearly outside the confidence envelope of the density function. This finding suggests that the original estimate is unlikely to be driven by random noise.

The placebo test strengthens the credibility of the causal inference by ruling out the possibility that the estimated treatment effect is confounded by latent trends specific to policy cities or sample heterogeneity. It further corroborates the baseline and event-study findings, indicating that the observed positive impact of the policy has substantive structural foundations.

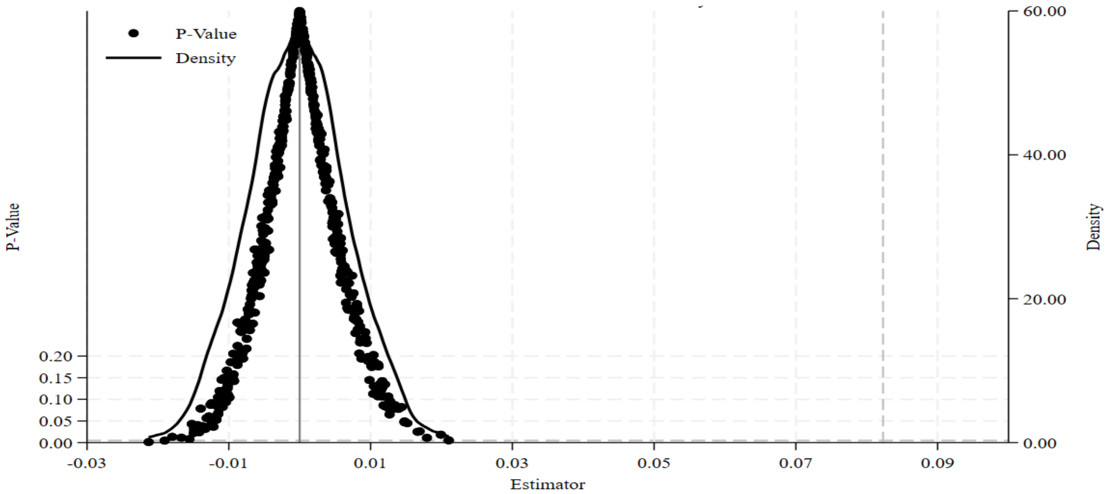

Figure 3. Placebo Distribution of Treatment Effects from Randomized Policy Assignment

Note: This figure displays the distribution of estimated treatment effects based on placebo replications using randomly assigned treatment groups and years. The x-axis represents the estimated coefficients, and the y-axis shows the associated p-values (left) and the kernel density of placebo estimates (right). The solid vertical line marks the actual coefficient from the baseline regression (β ≈ 0.08). The actual estimate lies in the far-right tail of the placebo distribution, indicating that the observed treatment effect is unlikely to result from random chance or sampling variability.

4. Robustness under Staggered Adoption: Goodman-Bacon Decomposition

To assess the robustness of the DID identification strategy under staggered policy implementation, this study further employs the Goodman-Bacon (2021) decomposition method to examine the composition of the overall estimator. As shown in Table 3, the dominant weight of the multi-period DID estimate (91.6%) originates from comparisons between treated units and never-treated units, with an associated average treatment effect of 0.043. The remaining weight comes from comparisons between groups with different treatment timings: the "earlier-treated vs. later-treated" group contributes 4.2% (average effect = 0.132), while the "later-treated vs. earlier-treated" group contributes 1.5% (average effect = 0.029). The overall estimate converges to 0.047, indicating that the primary result is almost entirely driven by treated vs. never-treated comparisons, mitigating concerns about bias from negative weights or timing-based distortions.

To further validate the findings, we employ the inverse probability-weighted estimator proposed by Callaway and Sant'Anna (2021), using untreated units as the control group. The estimated average treatment effect on the treated (ATT) is 0.047 (SE = 0.013, $p < 0.01$). This result remains highly consistent despite differences in estimation method, sample composition, and weighting structure compared to the baseline DID model, thereby reinforcing the stability and causal interpretability of the identified policy effect.

Table 3. Goodman-Bacon Decomposition

DD Comparison	Weight	Avg DD Est
Earlier T vs. Later C	0.042	0.132
Later T vs. Earlier C	0.015	0.029
T vs. Never treated	0.916	0.043
Diff-in-diff estimate	0.047	/

Note: This table presents the decomposition of the multi-period difference-in-differences estimator following Goodman-Bacon (2021). The overall DID estimate (0.047) is composed of weighted average comparisons between treated and untreated groups with varying treatment timing. The majority of the identifying variation (91.6%) comes from comparisons between treated units and never-treated units, with an average effect of 0.043. Comparisons between earlier and later treated units contribute only 4.2% and 1.5% of the weight, with estimated effects of 0.132 and 0.029, respectively. These results indicate that the average treatment effect is not driven by asynchronous treatment timing and is primarily identified from credible across-group contrasts.

5. Additional Robustness Checks

To further validate the stability of the identification results, we conduct a series of robustness tests from three dimensions. First, in Column (1), we exclude observations from 2008–2009 (the global financial crisis) and 2020–2021 (the COVID-19 pandemic), both representing periods of severe macroeconomic shocks. The estimated policy effect remains positive and statistically significant (coefficient = 0.084, SE = 0.029, $p < 0.01$), suggesting that the main findings are not driven by macro-level disruptions.

Second, Column (2) applies entropy balancing to reweight the control and treated groups, ensuring covariate balance before regression estimation. The estimated coefficient increases to 0.095 while retaining statistical significance, further enhancing the robustness of the results with respect to sample comparability.

Third, Column (3) uses year-by-year nearest-neighbor propensity score matching (1:3) to construct a matched sample and re-estimate the treatment effect. The resulting coefficient is 0.104 (SE = 0.030, $p < 0.01$), slightly larger than the baseline result. This suggests that the positive effect of the green data center policy on collaborative green innovation persists even in matched samples, effectively ruling out concerns about selection bias driving the results.

Table 4. Robustness Checks: Sample Restrictions, Weighting, and Matching

Variable	(1) Greenco	(2) Greenco	(3) Greenco
Policy	0.084***	0.095***	0.104***
	(0.029)	(0.029)	(0.030)
Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
City FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
N	10,641	13,754	10,837
Adj.R2	0.283	0.292	0.278

Note: This table reports robustness tests for the baseline policy effect on green collaborative innovation. Column (1) excludes years associated with the global financial crisis (2008–2009) and COVID-19 (2020–2021). Column (2) applies entropy balancing on key covariates (Size, Leverage, ROA, Cashflow, ListAge, Duality) prior to estimation. Column (3) uses a year-by-year 1:3 nearest-neighbor propensity score matching (PSM) approach with a caliper of 0.05. All specifications include firm-level controls and absorb firm, city, and year fixed effects. Standard errors are clustered at the firm level. *** $p < 0.01$.

5. Mechanism Analysis

The reconstruction of green collaborative innovation capacity relies not only on resource input but also on the coupling of internal informational capabilities, external network embeddedness, and organizational risk governance. The green digital policy, initiated through infrastructure-led institutional design, first manifests its effect by significantly enhancing firms' data disclosure capabilities (coefficient = 0.008, SE = 0.002). Unlike traditional compliance-based disclosures, this improvement stems from a cognitive shift within firms reframing previously informal and non-capitalized data resources into governable assets. Under policy mandates and external evaluative mechanisms (e.g., green catalogues and performance metrics), firms are compelled to respond by institutionalizing the assetization of data, laying a foundation for knowledge exchange and interface compatibility in green cooperation.

This cognitive elevation extends firms' external network boundaries. The regression results show that the policy significantly increases the likelihood of firms embedding into global innovation systems (coefficient = 0.080, SE = 0.024), particularly through overseas subsidiary formation and participation in international green patent collaborations. Such transboundary expansion reflects not merely market-driven behavior but a strategic response to enhanced institutional predictability. Green digital infrastructure reduces informational friction in external collaboration and improves firms' visibility as green innovation nodes. The resulting "connectivity" formed under institutional conditions serves as the external embedding basis for green collaborative innovation.

These mechanisms also generate structural effects on innovation risk. As collaborative R&D is often fraught with uncertainty, policy-induced stability and resource realignment significantly reduce firms' probability of innovation failure (coefficient = –0.076, SE = 0.024). This decline does not merely reflect risk aversion but rather a path-dependent adjustment grounded in updated expectations about institutional continuity and support availability. In this mechanism, the green digital policy does not act via direct subsidies but instead re-empowers high-risk strategic decisions through enhanced environmental certainty.

Table 5. Mechanism Tests: Data Disclosure, Innovation Networks, and Risk Reduction

Variable	(1) DataDisclosure	(2) GlInNet	(3) InnoRisk
Policy	0.008***	0.080***	-0.076***
	(0.002)	(0.024)	(0.024)
Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
City FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
N	13,754	13,754	13,754
Adj.R^2	0.139	0.773	0.206

Note: This table reports mechanism regressions examining how the green data center policy influences key channels underlying collaborative innovation. Column (1) uses a disclosure index based on firm-level data asset transparency; Column (2) indicates global innovation network embedding via foreign affiliate linkages; Column (3) captures innovation risk via a binary indicator of R&D-performance divergence. All specifications include firm-level controls and absorb firm, city, and year fixed effects. Standard errors are clustered at the firm level. *** $p < 0.01$.

6. Further Analysis

6.1 Moderation Effects

Figure 4 illustrates the heterogeneous marginal effects of green digital policies (GDPs) under different firm-level and industry-level conditions, revealing a structurally contingent moderation mechanism.

First, the interaction between data factor utilization and the policy variable is significantly positive (coefficient > 0, $p < 0.1$), indicating that policy incentives are more pronounced among firms that already possess capabilities in data identification and absorption. These firms typically have institutionalized data governance processes and cognitive foundations that allow them to translate green digital infrastructure into innovation opportunities. This suggests that the marginal productivity of institutional supply is not linearly increasing but exhibits threshold dependence on organizational cognitive capacity.

Second, firms with stronger green strategic orientation also experience amplified policy effects (coefficient > 0, $p < 0.1$). This finding implies that the effectiveness of institutional embedding relies critically on the presence of an internalized green vision. When green transformation is explicitly incorporated into a firm's governance language and strategic agenda, the firm is more likely to deliver targeted responses rather than treat the policy merely as a compliance obligation. In this sense, the triggering mechanism of policy incentives is highly conditional on the firm’s preexisting environmental cognition.

By contrast, the intensity of industry-level green competition appears to significantly dampen the policy's effect (coefficient < 0, $p < 0.1$). This suggests that in sectors where environmental certifications are already widespread and regulatory pressure is strong, the marginal impact of additional green digital policy is constrained. Two mechanisms may explain this: first, the marginal salience of new policy signals may weaken under institutional saturation; second, green certifications in highly regulated sectors may serve more symbolic compliance than genuine innovation, thereby diluting the intended effect of the policy.

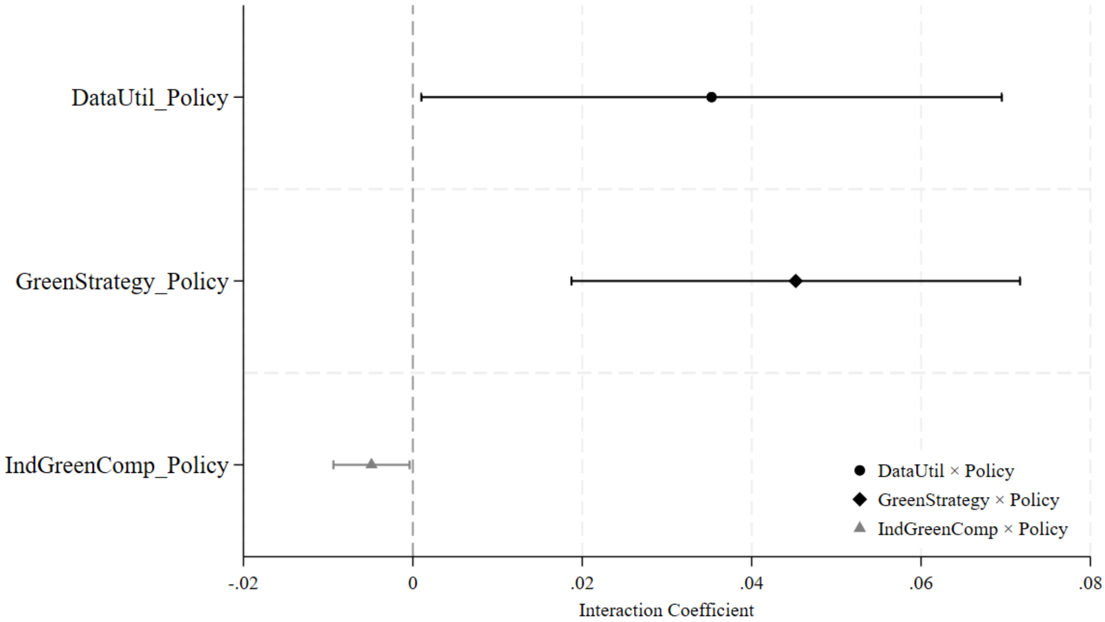

Figure 4. Moderating Effects of Firm Characteristics on the Policy Impact

Note: This figure presents interaction estimates between the green data center policy and three firm- or industry-level moderators. "DataUtil × Policy" reflects whether the firm possesses a high level of digital capability based on the use of AI, cloud, blockchain, and big data technologies. "GreenStrategy × Policy" indicates the presence of explicit green strategic commitments, including environmental goals and ISO14000 certification. "IndGreenComp × Policy" measures the intensity of green institutional competition within an industry based on the proportion of firms with ISO14001 certification. Interaction coefficients are estimated separately in fully controlled regressions with firm, city, and year fixed effects. The first two interactions are significantly positive ($p < 0.1$), while the third is significantly negative ($p < 0.1$), suggesting a suppressing effect of dense institutional environments.

4.2 Extensive and Intensive Margins

In the absence of institutionalized infrastructure support, firms' participation in green collaborative innovation is typically constrained by uncertainty, organizational capability gaps, and governance costs associated with cooperation. Through the construction of digital infrastructure and formal institutional certification mechanisms, green digital policies (GDPs) effectively reconfigure these three structural barriers.

Among firms with no prior engagement in green collaborative innovation, the policy acts as a signal that enhances the perceived safety of participation, thereby significantly increasing the likelihood of initial entry into green cooperation (coefficient = 0.043, $p < 0.05$). This triggering effect is not contingent on the presence of existing collaborative paths but rather emerges from the institutional design's ability to reshape firms' perception of actionability under risk.

For firms already involved in collaborative green innovation, the effect of the policy manifests as a deeper, more cohesive extension of existing efforts. Institutional supply stabilizes trust mechanisms in repeated strategic interactions and provides more granular data interfaces and compliance frameworks for sustained cooperation. As a result, not only is green collaboration maintained, but it is also significantly intensified under a more robust institutional environment (coefficient = 0.503, $p < 0.01$). The observed increase in the intensive margin does not stem from a simple amplification of incentives but from a revaluation of the "controllability of returns" associated with sustained cooperation under institutional certainty.

Importantly, the extensive and intensive margins are not dichotomous. Rather, they reflect a dynamic reallocation of organizational response intensity as institutional clarity increases. GDPs do not directly produce innovation outputs; instead, they construct a new structural mapping between behavioral thresholds, collaborative density, and organizational expectations.

Table 6. Extensive vs. Intensive Margins of Green Collaborative Innovation

Variable	(1) Greenco_dummy	(2) Greenco
Policy	0.043**	0.503***
	(0.018)	(0.170)
Controls	Yes	Yes
Firm FE	Yes	Yes
City FE	Yes	Yes
Year FE	Yes	Yes
N	13,754	599
Adj.R2	0.247	0.096

Note: This table separates the effect of the green data center policy into extensive and intensive margins. Column (1) uses a dummy variable indicating whether the firm filed any green collaborative patent in a given year. Column (2) uses the log count of joint green patent applications, restricting the sample to observations with positive values. The results show that the policy significantly increases both the likelihood of engaging in green collaboration ($p < 0.05$) and the intensity of such collaboration conditional on participation ($p < 0.01$). All regressions control for firm-level covariates and absorb firm, city, and year fixed effects. Standard errors are clustered at the firm level.

4.3 Patent-Type Heterogeneity

The output forms of green collaborative innovation exhibit substantial heterogeneity across patent categories. Invention patents and utility model patents differ fundamentally in their underlying knowledge coordination mechanisms, intellectual property delineation, and organizational complexity of collaboration. As shown in Table 7, GDPs exert significantly positive effects on both types of patents, indicating that institutional incentives are effective across varying levels of innovation complexity.

Although the regression coefficients cannot be statistically compared across models, the divergence in underlying mechanisms remains analytically meaningful. Invention-type green patents typically involve greater degrees of technological integration and inter-organizational knowledge sharing, making their realization more reliant on institutional environments that facilitate information fluidity and clarify property rights. GDPs, by enhancing information transparency and lowering coordination costs, offer institutional soil conducive to such high-complexity collaborations.

In contrast, utility model patents tend to focus on incremental improvements and low-barrier cooperation. While less dependent on institutional sophistication, these innovations still benefit from the policy-induced improvements in resource allocation and reinforcement of green signaling. The results suggest that even under relatively low-complexity conditions, institutional interventions can generate substantial incentive effects.

These findings uncover the compatibility mechanism of GDPs across multi-tiered innovation contexts. On the one hand, they provide governance support for complex cooperation; on the other, they deliver foundational incentives for widespread, lower-threshold innovation activities. This reflects the broad-spectrum adaptability of institutional tools in promoting green innovation.

Table 7. Policy Effects on Different Types of Green Collaborative Patents

Variable	(1) G_innov	(2) G_utility
Policy	0.062***	0.046**
	(0.022)	(0.020)
Controls	Yes	Yes
Firm FE	Yes	Yes
City FE	Yes	Yes
Year FE	Yes	Yes
N	13,754	13,754
Adj.R2	0.241	0.206

Note: This table presents separate estimates of the green data center policy's impact on two types of green collaborative patents. Column (1) uses the count of jointly filed green invention patents, which typically involve higher technological complexity and knowledge integration. Column (2) uses the count of jointly filed green utility model patents, which generally reflect incremental improvements. The policy shows statistically significant positive effects on both types of innovation. All regressions control for firm-level characteristics and include firm, city, and year fixed effects. Standard errors are clustered at the firm level.

7. Conclusion and Policy Implications

7.1 Conclusion

The positive effect of GDPs on the green collaborative innovation of HPFs is empirically verified, while such an effect is not observed among non-polluting firms. This finding suggests that the effectiveness of GDPs is not a universal property of policy tools, but rather highly contingent on the structural constraints of the organizational environment. Institutional induction of green collaboration does not operate directly; instead, it functions through three interrelated capability channels: enhanced data asset disclosure improves firms' visibility into green risks and opportunities, embeddedness in global innovation networks expands the accessible boundary of external knowledge, and reduced innovation failure risk stabilizes behavioral expectations under uncertainty.

The boundary of policy effectiveness is further revealed through firm-level heterogeneity. The ability to utilize data factors and the presence of green strategic orientation jointly constitute an internal absorptive mechanism for policy signals, determining whether institutional supply can be effectively translated into organizational behavior. In contrast, in industries with dense green regulatory coverage, GDPs exhibit a clear attenuation tendency suggesting that regulatory saturation may weaken firms' behavioral sensitivity to marginal policy changes.

Moreover, the impact of GDPs is not limited to facilitating firm entry into green collaborative networks. The significant effect observed at the intensive margin implies that firms not only decide whether to act, but also adjust the intensity of their collaborative behaviors. The absence of directional differences in the response between green invention patents and utility models supports the view that policy incentives operate in a path-neutral and complexity-agnostic manner.

Overall, the efficacy of GDPs stems from their ability to reconfigure firms' cognitive frameworks and capability boundaries, rather than from direct resource allocation or coercive mandates. Among firms characterized by high pollution intensity, well-defined internal capabilities, and clear strategic orientations, policy signals are more likely to be absorbed and converted into collaborative action—thus forming an endogenous coupling between institutional supply and organizational response. Further analysis shows that GDPs influence not only whether firms participate in green collaborative innovation, but also how intensely they engage within existing partnerships. This dual-margin effect persists across technical paths of varying complexity, highlighting the institutional depth of GDPs in both the initiation and amplification of green collaborative behavior.

4.2 Policy Implications

The institutional effectiveness of green collaborative innovation is not inherently guaranteed; rather, it emerges from a coupling process involving digital governance logics, organizational capability structures, and policy recognition capacity. Within this framework, GDPs do not operate primarily through direct resource infusion or coercive regulatory pressure, but rather through "translation pathways" formed by organizational absorption, cognitive reinterpretation, and capability deployment. Hence, the sustainability of policy outcomes hinges less on the magnitude of policy diffusion and more on whether firms possess the internal structure to transform institutional input into coordinated action.

From the supply-side perspective, the incentive foundation of GDPs lies in their institutionalized digital infrastructure, not merely in their physical deployment. The integration of “green” and “digital” under GDPs is not about energy-efficiency constraints or hardware specifications, but about the extent to which digital governance mechanisms are in place to support risk identification, collaborative search, and resource optimization. Accordingly, the design of GDPs should prioritize embedding such governance logic within the internal decision-making processes of firms especially HPFs so that data assets and green cognition are internalized as components of their incentive architecture. Only when data factors are woven into firms’ behavioral logic can their collaborative capacities and risk sensitivities be institutionally responsive.

At the same time, the effectiveness of GDPs is shown to be highly non-linear under conditions of organizational heterogeneity and regulatory density. Among firms lacking sufficient absorptive capacity for data elements or a clearly defined green strategy, even strong institutional signals are unlikely to trigger behavioral change. Conversely, in industries already saturated with green regulatory requirements, additional policies may dilute incentives or blur institutional boundaries. This suggests that the core policy challenge is not simply "whether to intervene", but "who is capable of receiving," "who can translate," and "who is willing to respond." Institutional signals must be embedded in organizational structures that possess both cognitive readiness and strategic willingness, or risk becoming silent or drowned out.

Thus, optimizing GDPs should not focus solely on expanding external coverage but on enhancing the alignment among policy supply, governance interfaces, and capability thresholds. Effective policy design requires attention to organizational compatibility: improving institutional visibility, enhancing organizational responsiveness, and reinforcing behavioral stability. Specifically, efforts to standardize data disclosure, increase transparency in green collaborations, and introduce risk-governance instruments for innovation failure are essential to constructing a coherent policy–capability–behavior chain. Only through such alignment can GDPs simultaneously stimulate the willingness and the capacity of HPFs to undertake green transformation.

4.3 Limitations

While the institutional effects of GDPs are identified among HPFs, firms’ behavioral responses to such policies may not be solely determined by whether they are located in pilot cities. The potential desynchronization between formal policy coverage and actual firm-level perception raises concerns regarding the spatial misalignment of institutional exposure. Thus, identifying treatment solely based on administrative boundaries may introduce measurement noise and obscure the true penetration of policy signals.

Moreover, green collaborative innovation—being an inherently inter-organizational process—entails complex cognitive negotiations and contractual arrangements, which are difficult to capture through the structural proxies used in this study. While our mechanism tests employ theoretically grounded variables, they are ultimately based on indirect behavioral indicators and may fall short in uncovering the internal governance logic or the dynamic formation of cooperative agreements within firms.

The staggered implementation of GDPs across regions and the multi-stage nature of their rollout also complicate causal inference. Although the study adopts a multi-period DID framework to mitigate time heterogeneity, policy enactment often coincides with local supporting measures, shifting government expectations, or evolving market competition, potentially introducing compound exposures that cannot be fully disentangled. Additionally, the mechanism analysis—despite incorporating three theoretically relevant variables—relies on publicly available disclosures and textual data, limiting the capacity to observe firm-level knowledge management, incentive design, and strategic interplay in real time. The absence of high-frequency institutional documents or interactive micro-level datasets constrains the granularity with which institutional perception and response processes can be captured.

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